

Lecture 11 – Wind Energy

Part I: Wind Turbine and Energy Conversion

Dr. Yu Zhang

ECE, UCSC

ECE180J

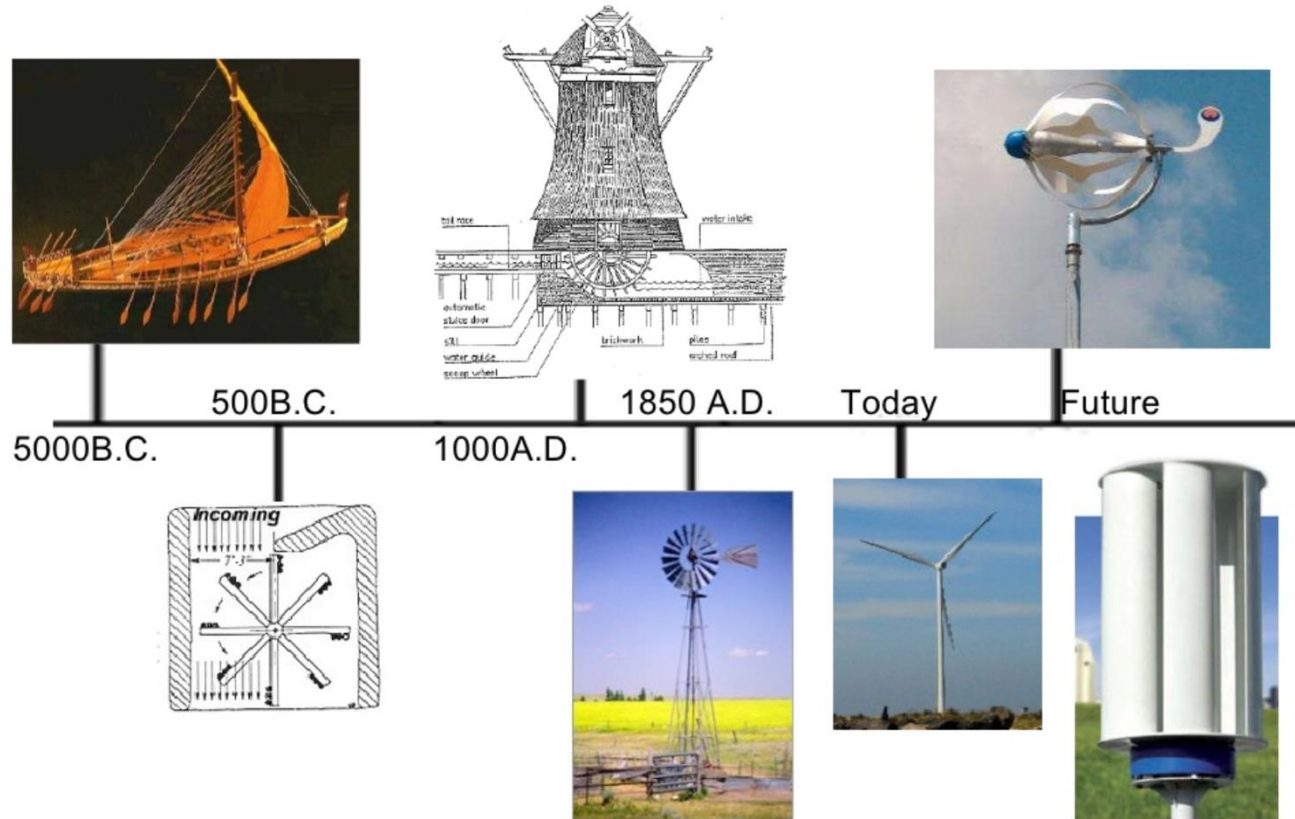


Outline

- History and Development of Wind Energy
 - Wind Turbine Structure and Types
 - Harnessing Aerodynamic Forces
 - Onshore and Offshore Turbines
 - Wind Turbine Costs
 - Environmental Impact
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- B3 Chap 8
 - B1 Chap 7

History of Wind Power

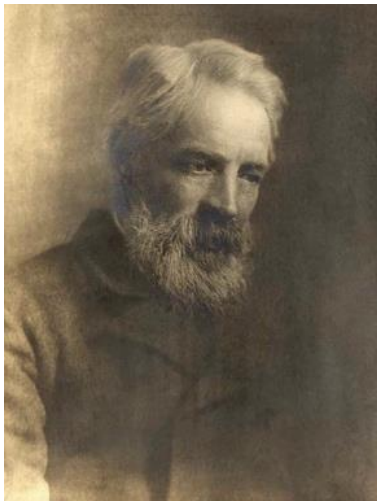
- Wind energy has been used for thousands of years for milling grain, pumping water and other mechanical power applications.



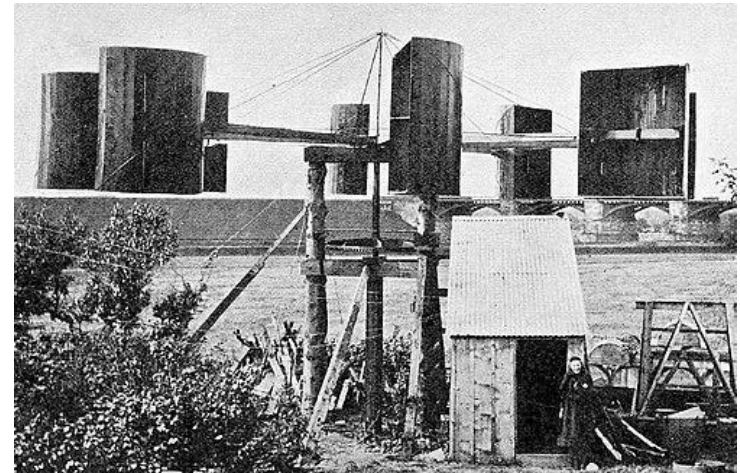
1. Wind energy propelled boats along the Nile River as early as 5000 B.C.
2. Simple windmills in China were pumping water,
3. Vertical-axis windmills were grinding grain in the Middle East.

Pioneer 1: James Blyth in Scotland

- In 1887, James Blyth built a cloth-sailed wind turbine in his holiday cottage, and used the produced electricity to charge accumulators.
- Stored electricity further powered the lights, which made the cottage become the **first house in the world to be powered by wind-generated electricity.**



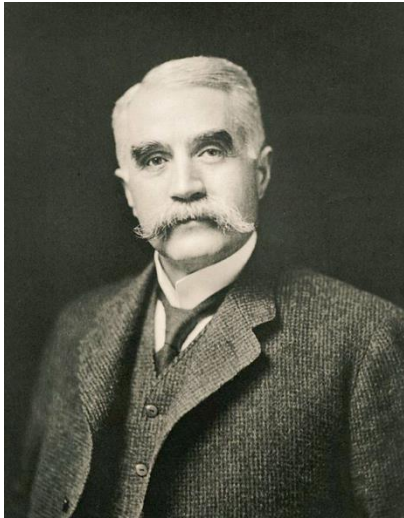
James Blyth (1839-1906): Scottish electrical engineer and a professor



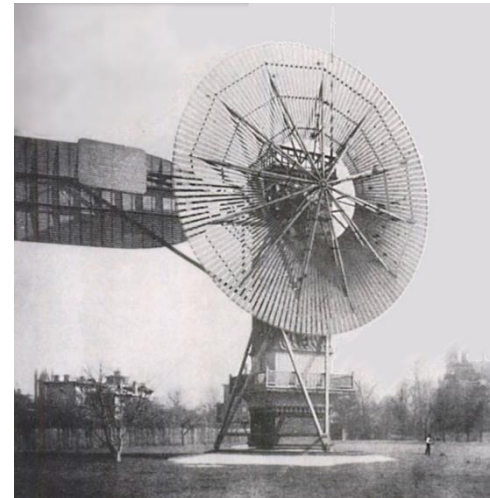
Blyth's windmill at his cottage in Marykirk: a vertical axle and cup-like structures.

Pioneer 2: Charles Brush in the US

- In 1888, Charles Brush powered his mansion with the **world's first automatically operated wind turbine generator**.
- It charged the home's 12 batteries. It was the first home in Cleveland to have electricity.
- Over its 20-year life, the turbine never failed to keep the home continuously powered.



Charles Brush (1849 –1929), American engineer, inventor, entrepreneur, & philanthropist.



The world's first automatically operated wind turbine was built in 1888. It had a 12kW dynamo.

Pioneer 3: Poul la Cour in Denmark

- Poul la Cour was a pioneer in the field of aerodynamics and is known for his development of windmills for electricity generation.
- Developed the theory for the ideal windmill blade profile.
- Studied the potential of storing wind energy in hydrogen & oxygen by electrolysis.



Poul la Cour (1846 – 1908): Danish scientist and inventor



The Test Turbines in Askov - 1897

Development of Wind Power



FIGURE W-1

Wind Power Global Capacity and Annual Additions, 2015-2024

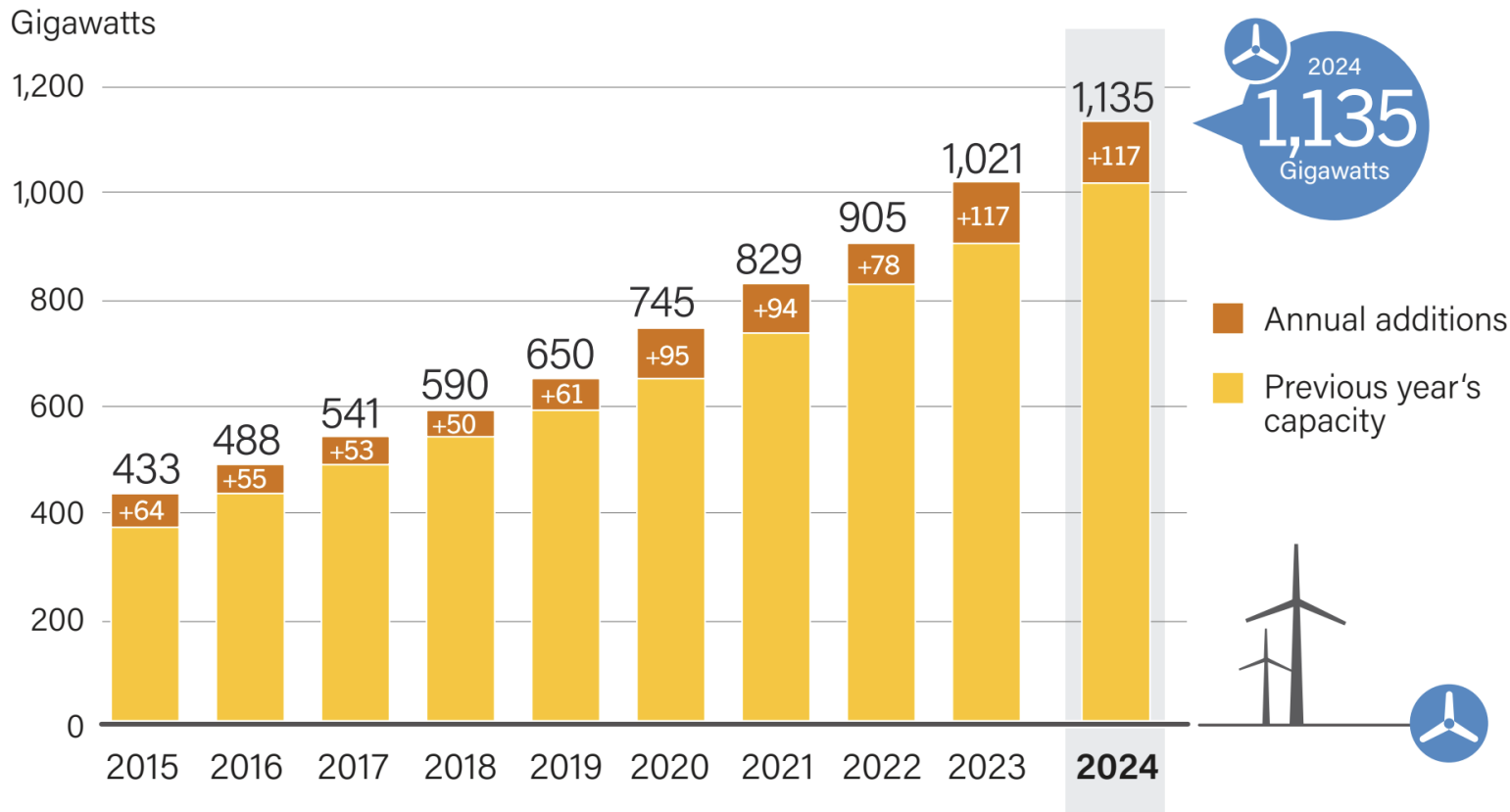


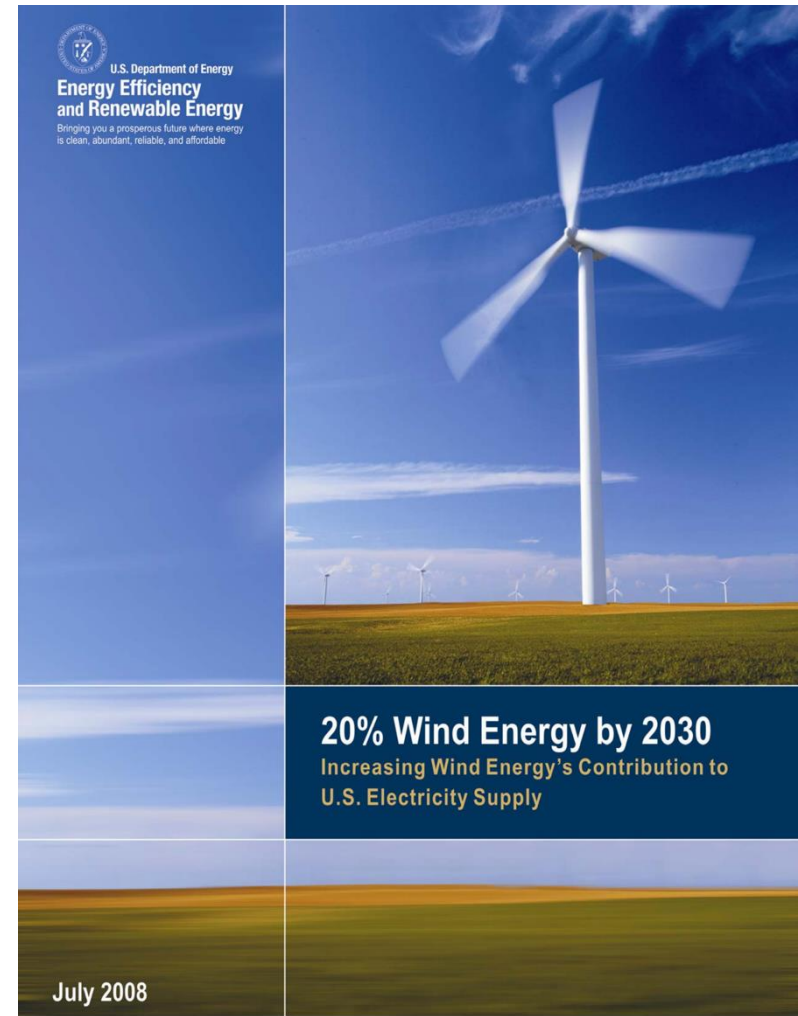
Figure Source: Ren21 Renewables 2025 Global Status Report

https://www.ren21.net/wp-content/uploads/2019/05/GSR2025_Factsheet_Supply_Wind.pdf

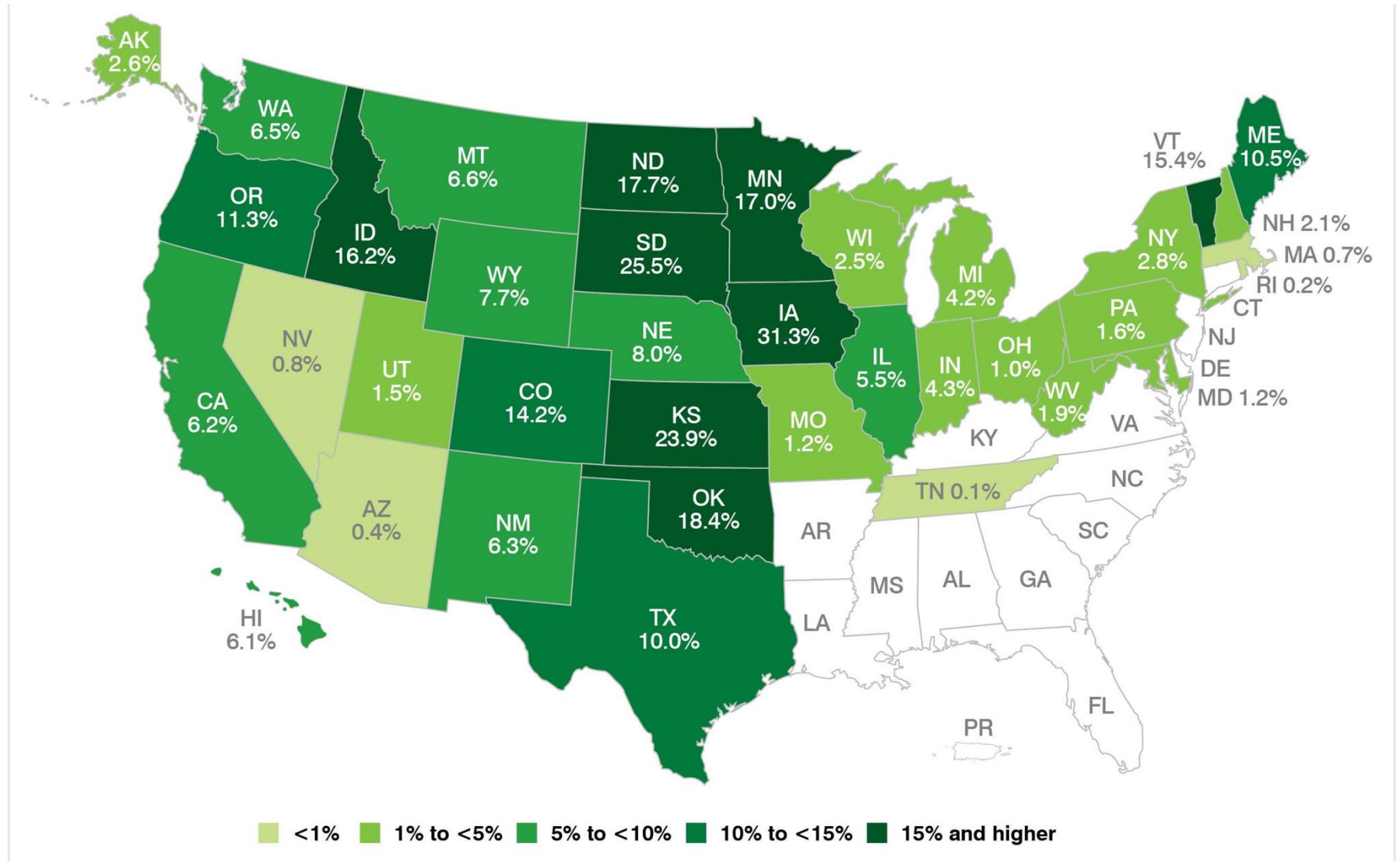
Goal

U.S. DoE: Wind energy supplying 20% of the country's electricity in 2030, and 35% in 2050.

- 1) require enhanced transmission infrastructure, streamlined siting and permitting regimes.
- 2) require the number of turbine installations to increase from approximately 7000/year in 2017.
- 3) done reliably for less than 0.5 cents per kWh.
- 4) not limited by the availability of raw materials.



Wind Energy Share of Electricity Generation in 2015

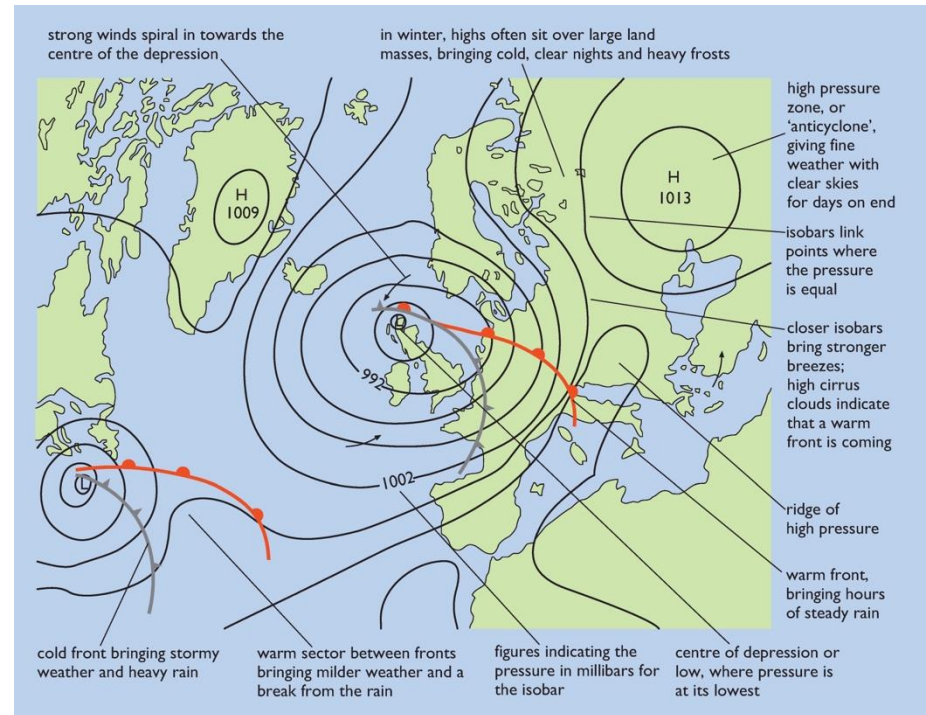
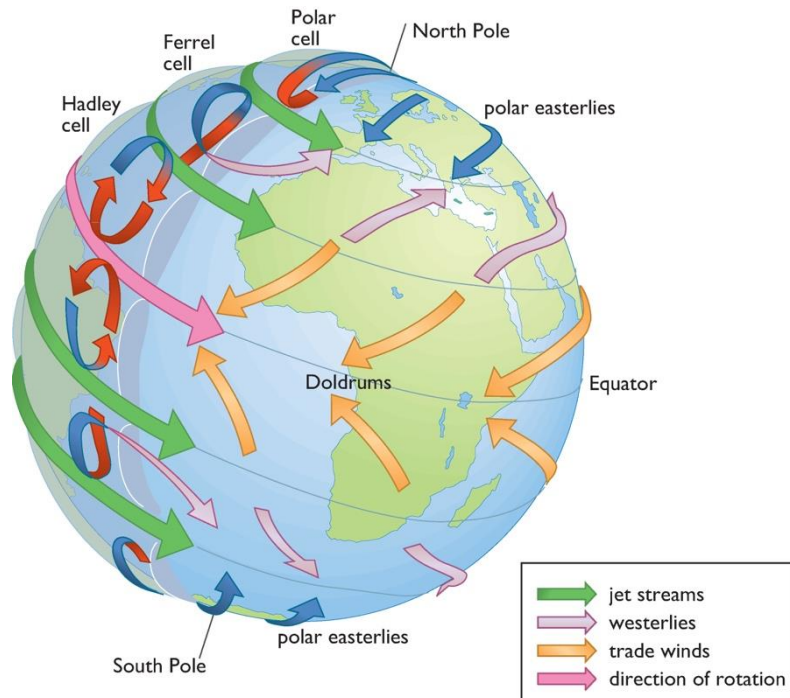


The Origins of Wind

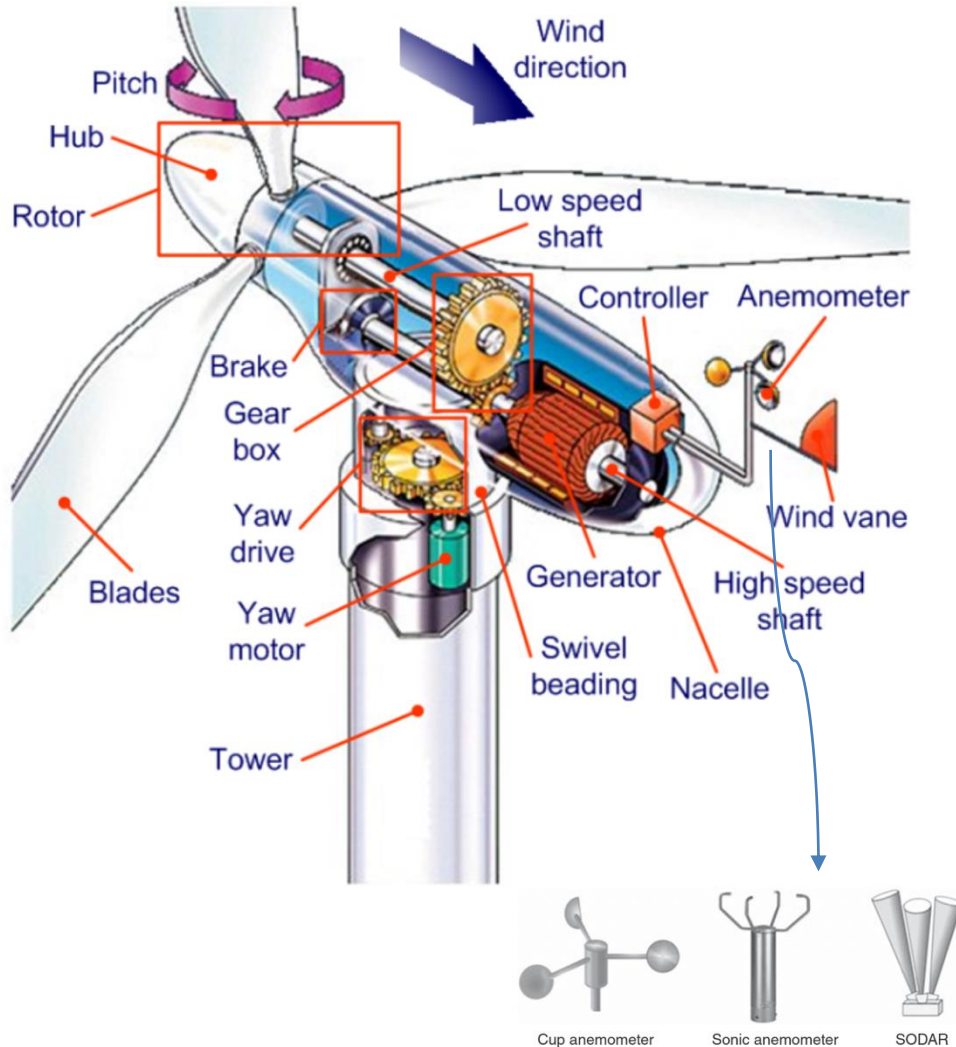
Differential solar heating of the Earth's surface causes **variations in atmospheric pressure** → movements of air masses → wind.

Weight of the column of air above a specified surface area yields atmospheric pressure (unit: 1 bar = 100,000 Pa).

The average atmospheric pressure at sea level is about 1 bar.

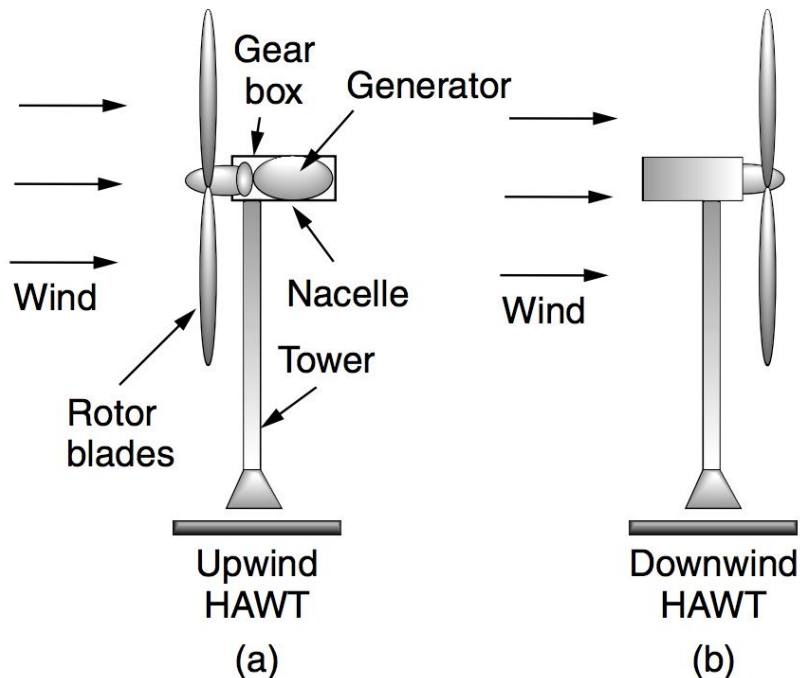


Wind Turbine Structure



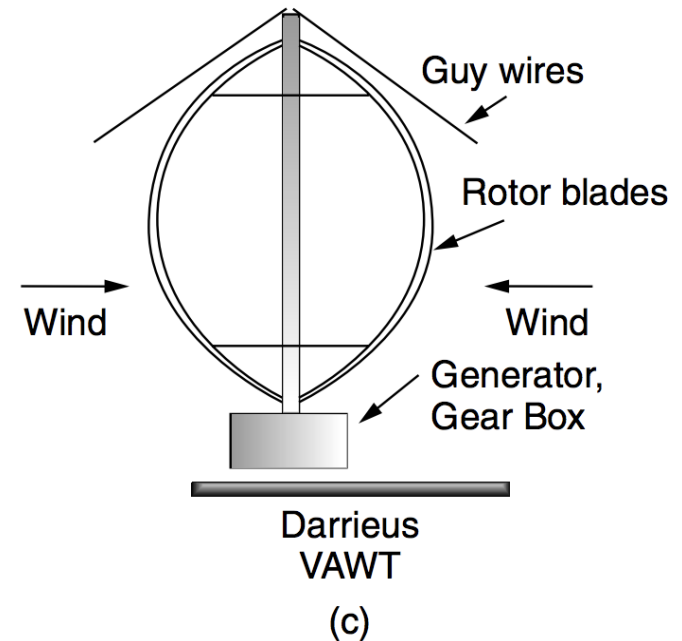
- Blades are used to convert kinetic energy in the wind into rotating shaft power to spin a generator that produces electric power.
- A gearbox transfers power from the low speed shaft to a high speed shaft that spins the generator.
- Yaw motor keeps the blades facing into the wind.
- Brake engages to lock the blades in place when winds are too strong.

Wind Turbine Types: HAWT and VAWT



Horizontal axis wind turbines (HAWTs):
rotation axis is in line with the wind direction

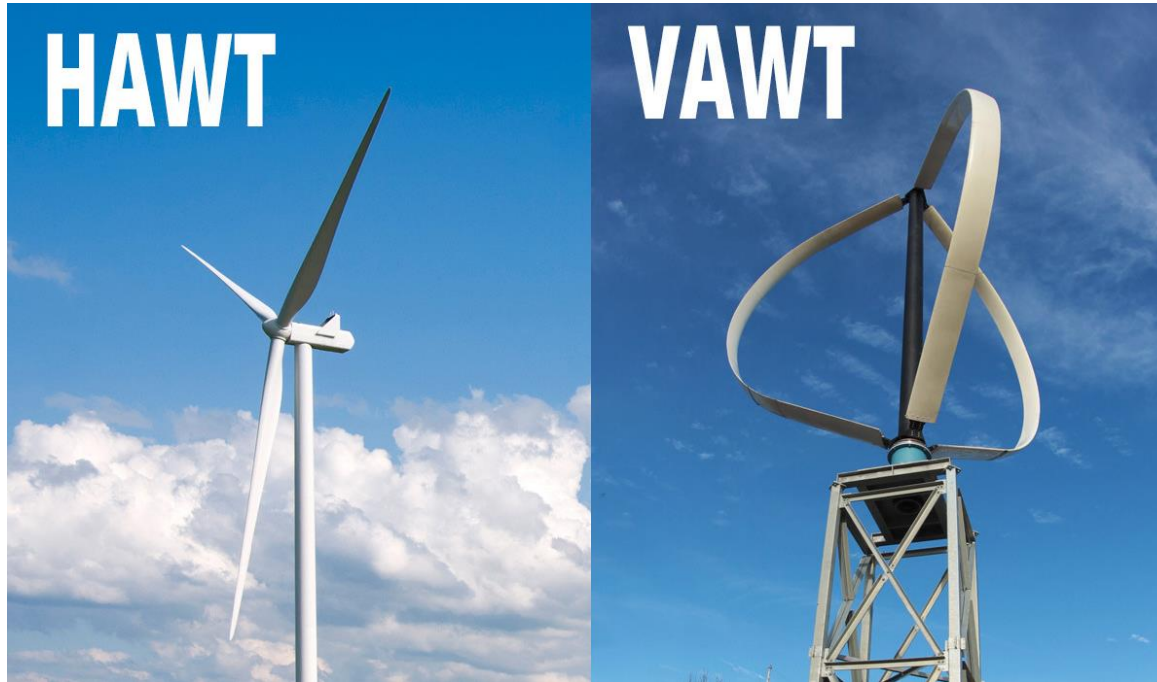
- Downwind HWAT naturally let wind control the yaw, but have the flexing issue leading to blade failure, increased noise, and reduced power output.



Vertical axis wind turbines (VAWT):
rotor shaft is set transverse to the wind

- Heavy generator and gear box are located down on the ground.
- little commercial success, due to issues with power quality, cyclic loads and the lower efficiency designs.

HAWT vs VAWT



HAWT

- Higher efficiency
- Dominates industry
- Needs yaw control
- Large commercial success

VAWT

- Easier maintenance
- Generator near ground
- Works from any direction
- Lower efficiency

Why Three Blades?

Three-blades design is a compromise between

- high energy yield
 - greater stability and durability
-
- A single blade minimizes drag and maximizes energy yield, but it causes balance issues.
 - Two-bladed turbines are susceptible to gyroscopic precession, leading to wobbling and reduced stability.
 - More than three blades increase wind resistance, reducing rotational speed and overall efficiency.

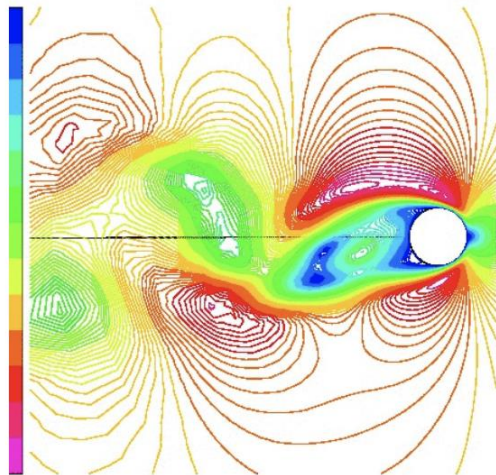


Bladeless Wind Turbines



Fig: Vortex Bladeless Wind Generator

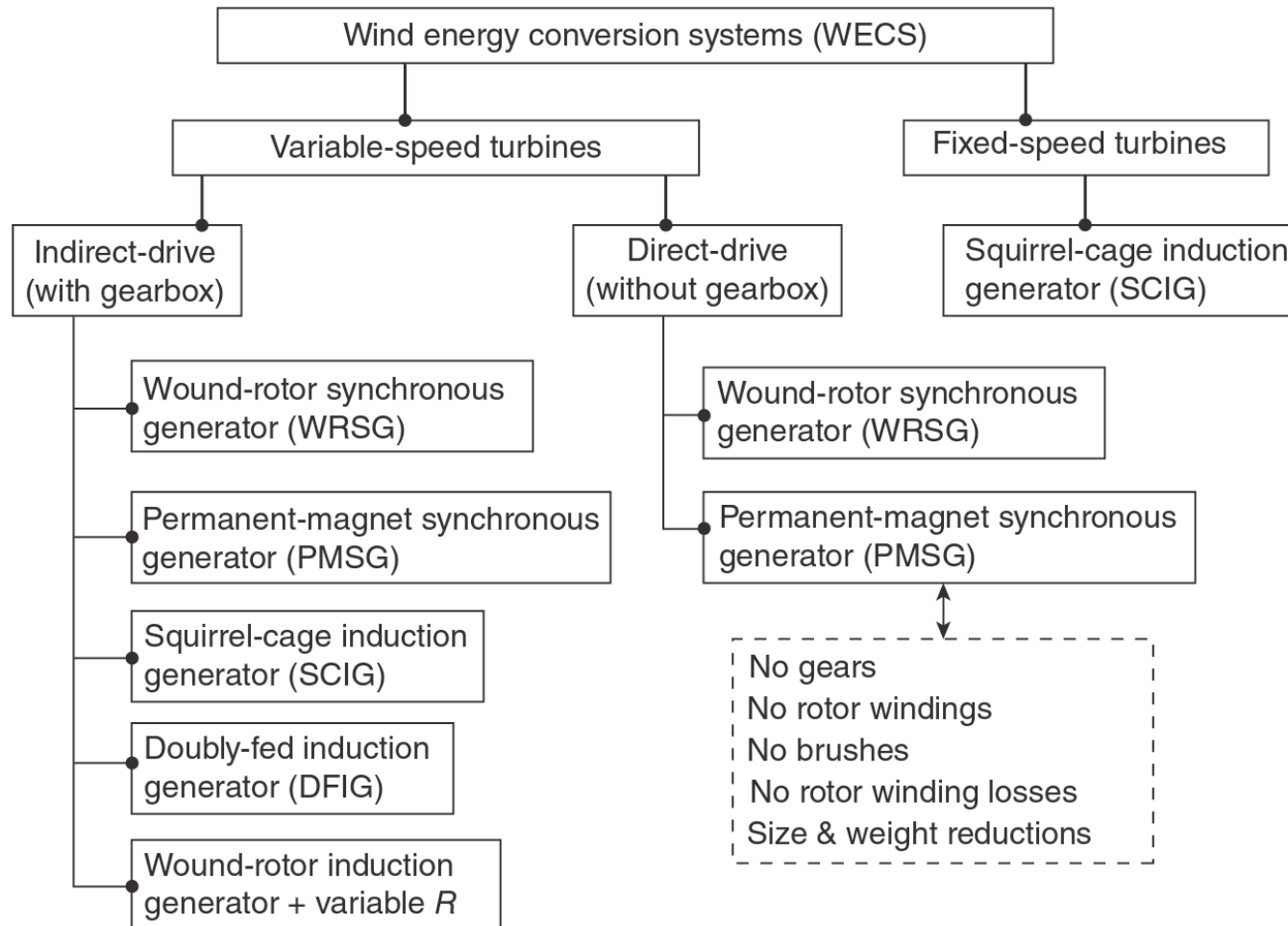
- Consist of a **cylinder** fixed vertically with an **elastic rod**.
- The cylinder oscillates on a wind range, which then generates electricity through an alternator system.



- Fluid mechanics: as the wind passes through a blunt body, the flow is modified and generates a cyclical pattern of vortices.
- Once the frequency of these forces is close enough to body's structural frequency, the body starts to oscillate and enters into **resonance** with the wind (a.k.a. Vortex Induced Vibration).

Wind Energy Conversion Systems (WECS)

- Variable-speed or fixed-speed rotor (here means rotor speed)
- Synchronous generator or induction generator
- With or without gearbox



Synchronous Generator

Synchronous generators (great majority in power systems)

- can supply reactive power for voltage control
- use permanent magnets or field windings to create the magnetic field
- rotor speed N and electrical frequency f are synchronized

$$N(\text{rpm}) = \frac{120f}{p}$$

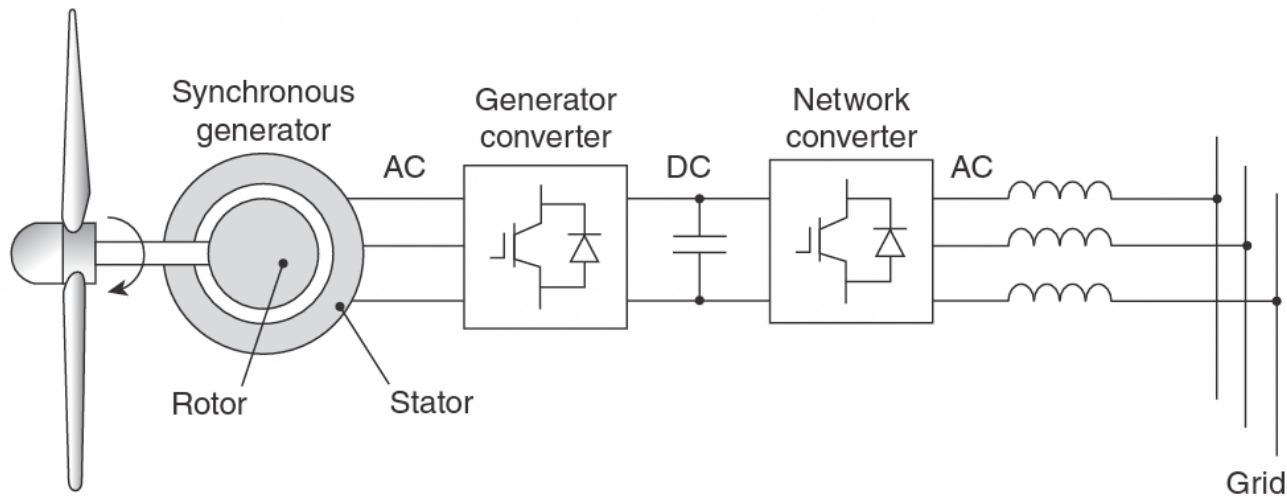


FIGURE 7.13 A gearless, variable-speed synchronous generator with full-capacity converters.

Induction Generator

Induction generators (used by most of the world's wind turbines)

- require grid/reactive power for magnetization
- consume reactive power
- rotor speed differs slightly from synchronous speed (slip)
- also called asynchronous generators

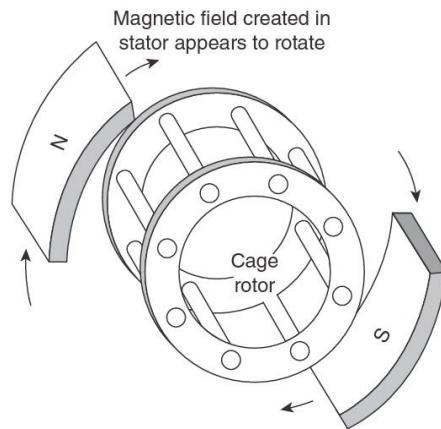


FIGURE 7.10 A cage rotor consists of thick, conducting bars shorted at their ends, around which circulates a rotating magnetic field.

- **Squirrel-cage rotor:** no need of exciter, brushes, and slip rings (simple & robust)
- Pole switching: remotely switch the number of poles by clever wiring of the stator windings.
- Slip speed: see details on page 402 of B1.

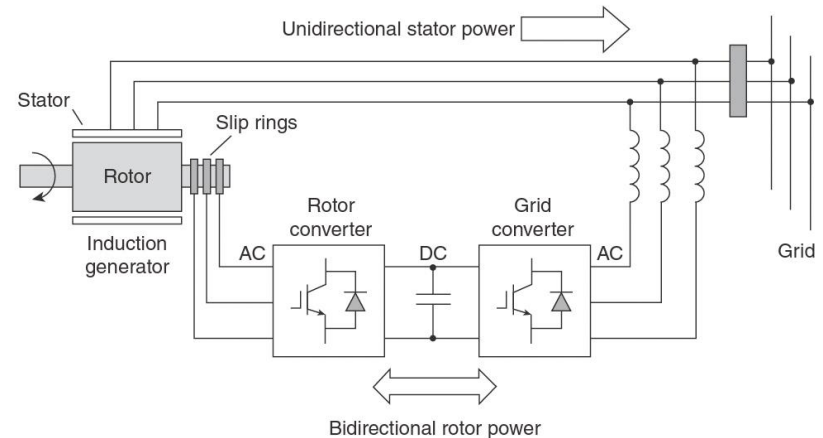


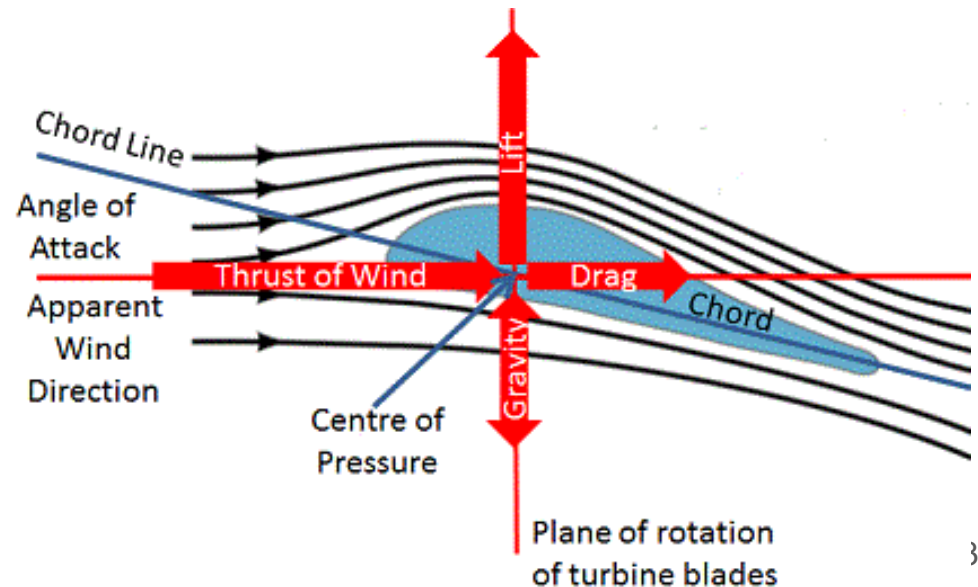
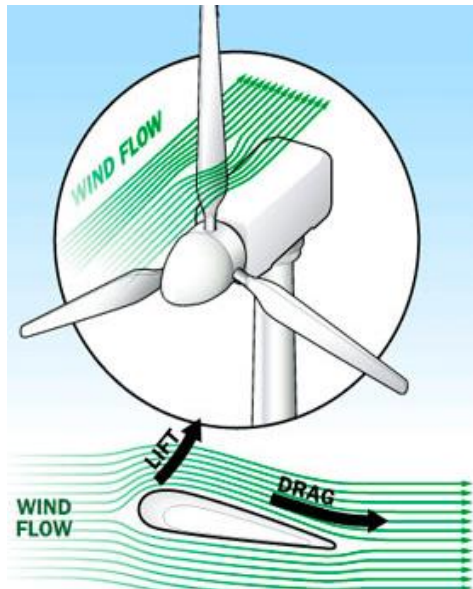
FIGURE 7.12 A wound-rotor, doubly-fed induction generator (DFIG).

- **Wound rotor:** the grid creates the stator's rotating magnetic field.
- Slip rings are used to energize the rotor.
- Bidirectional rotor power: power always flows from high freq side to low freq side.

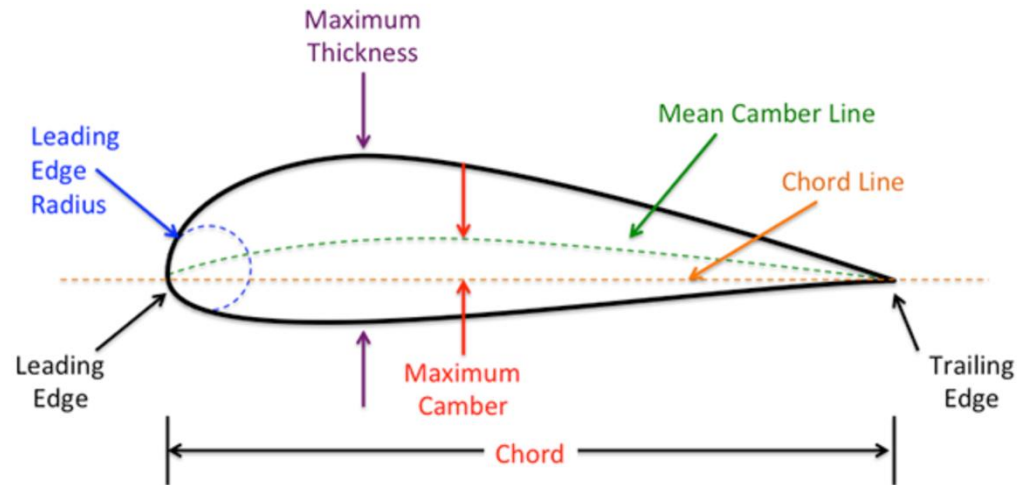
From Aircraft to Wind Turbine



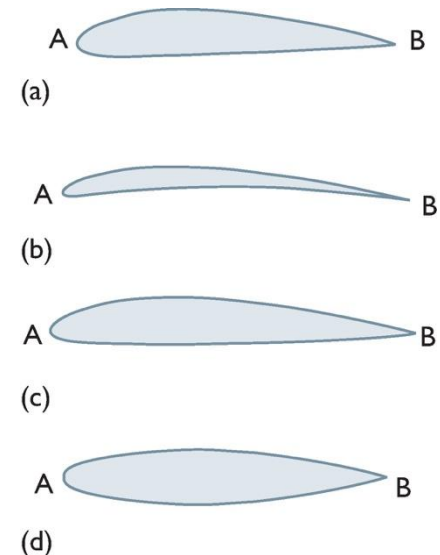
The faster the airflow, the lower the pressure
(the greater the 'suction effect')



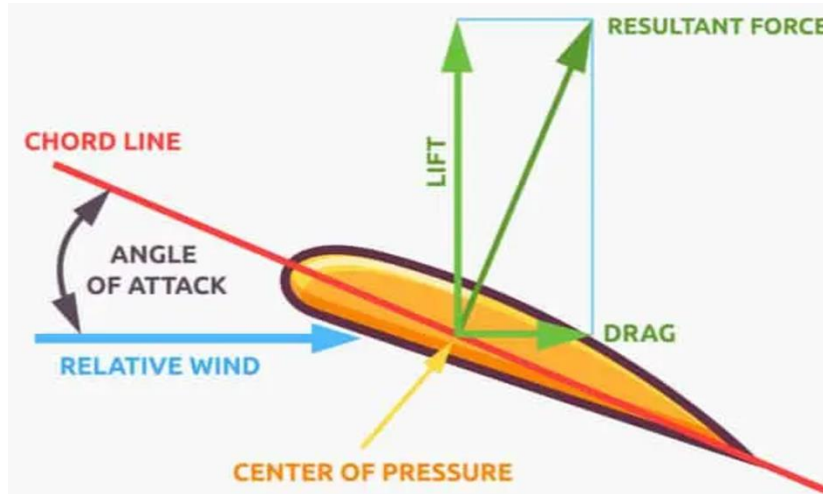
Airfoil Shape



- Airfoils are streamlined and optimized for different operating speeds.
- **Asymmetrical airfoils (a) – (c)**: Optimized to produce most lift when the underside of the airfoil is closest to the direction from which the air is flowing.
- **Symmetrical airfoils (d)**: Able to induce lift equally well when the air flow is approaching from either side of the 'chord line'.

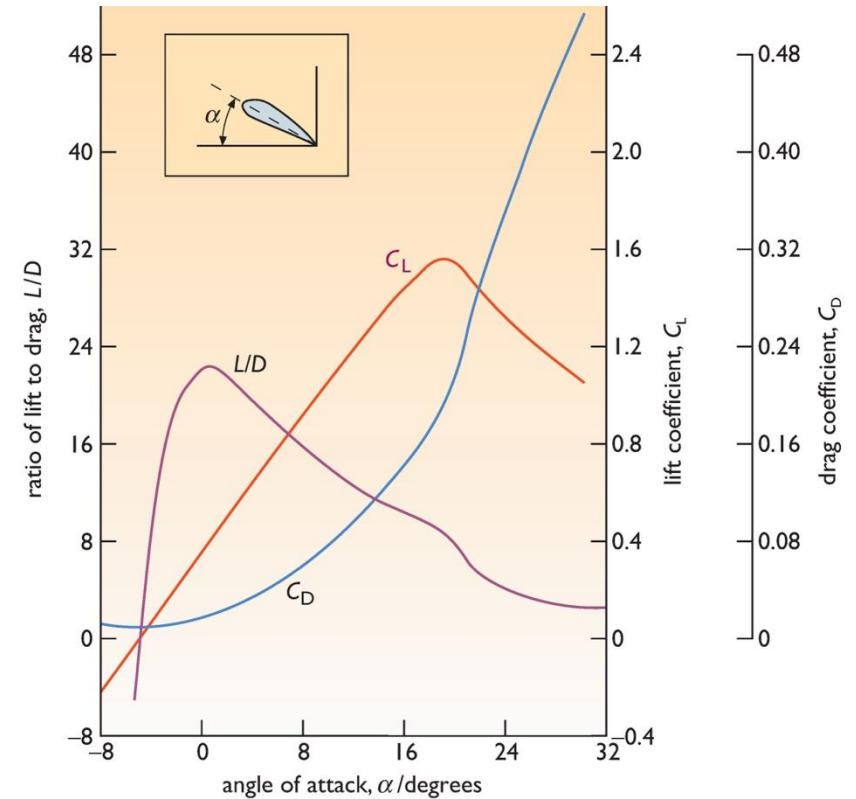


Lift and Drag Coefficients



$$\text{Drag coefficient : } C_D = \frac{D}{0.5\rho V^2 A_b}$$

$$\text{Lift coefficient : } C_L = \frac{L}{0.5\rho V^2 A_b}$$



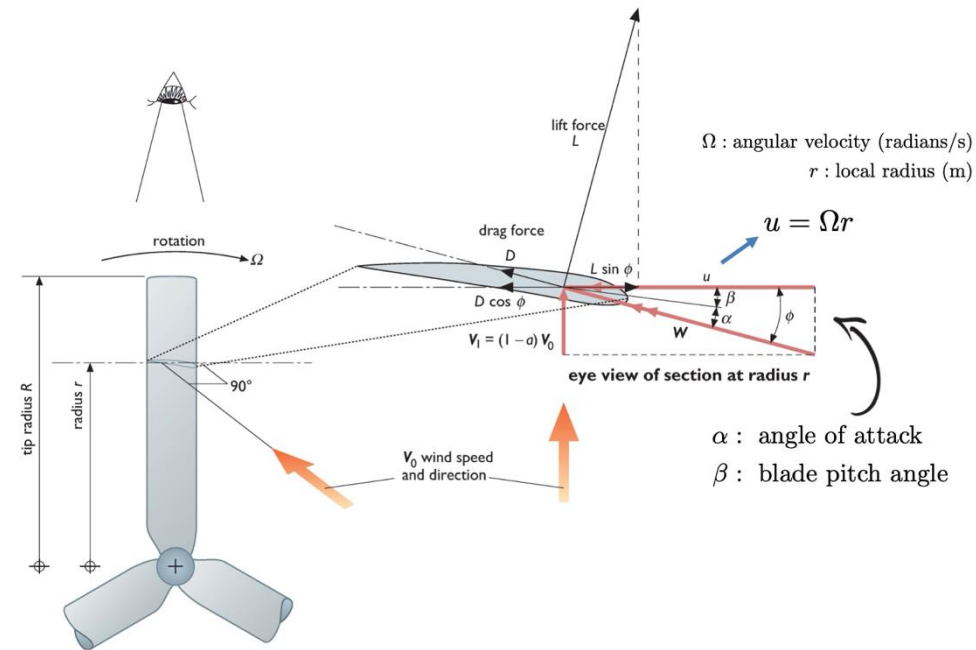
- $\frac{1}{2}\rho V^2$: dynamic pressure of the airflow
- A_b : reference area (typically blade/swept area)

Harnessing Aerodynamic Forces

HAWT & VAWT harness aerodynamic forces in a different way:

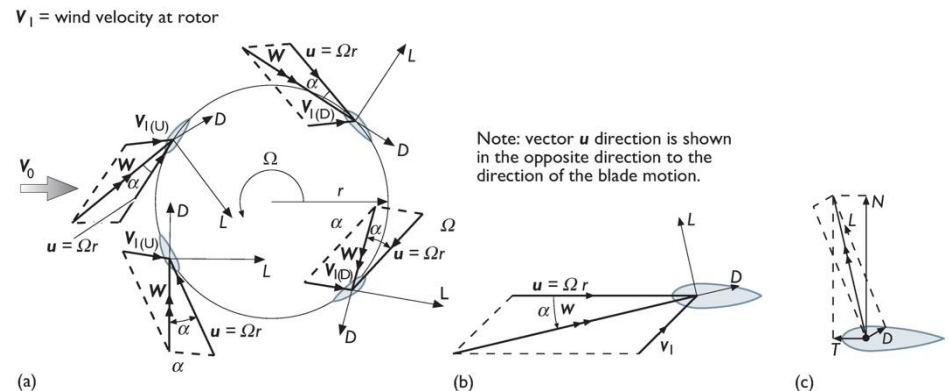
HAWT

- Nearly constant angle of attack
- Higher efficiency
- Uses asymmetric airfoils



VAWT

- Continuously varying angle of attack
- Lower efficiency
- Often uses symmetrical airfoils



Onshore or Offshore?

Pros:

- An offshore turbine can last around 25-30 years
- Produce about 50% more energy (3-5MW per hour w/ strong wind)
- Higher and more constant wind speed

Cons:

- More expensive to built
- More difficult to maintain & access



Onshore wind turbines



Offshore wind turbines

Foundation Tech (1/3)

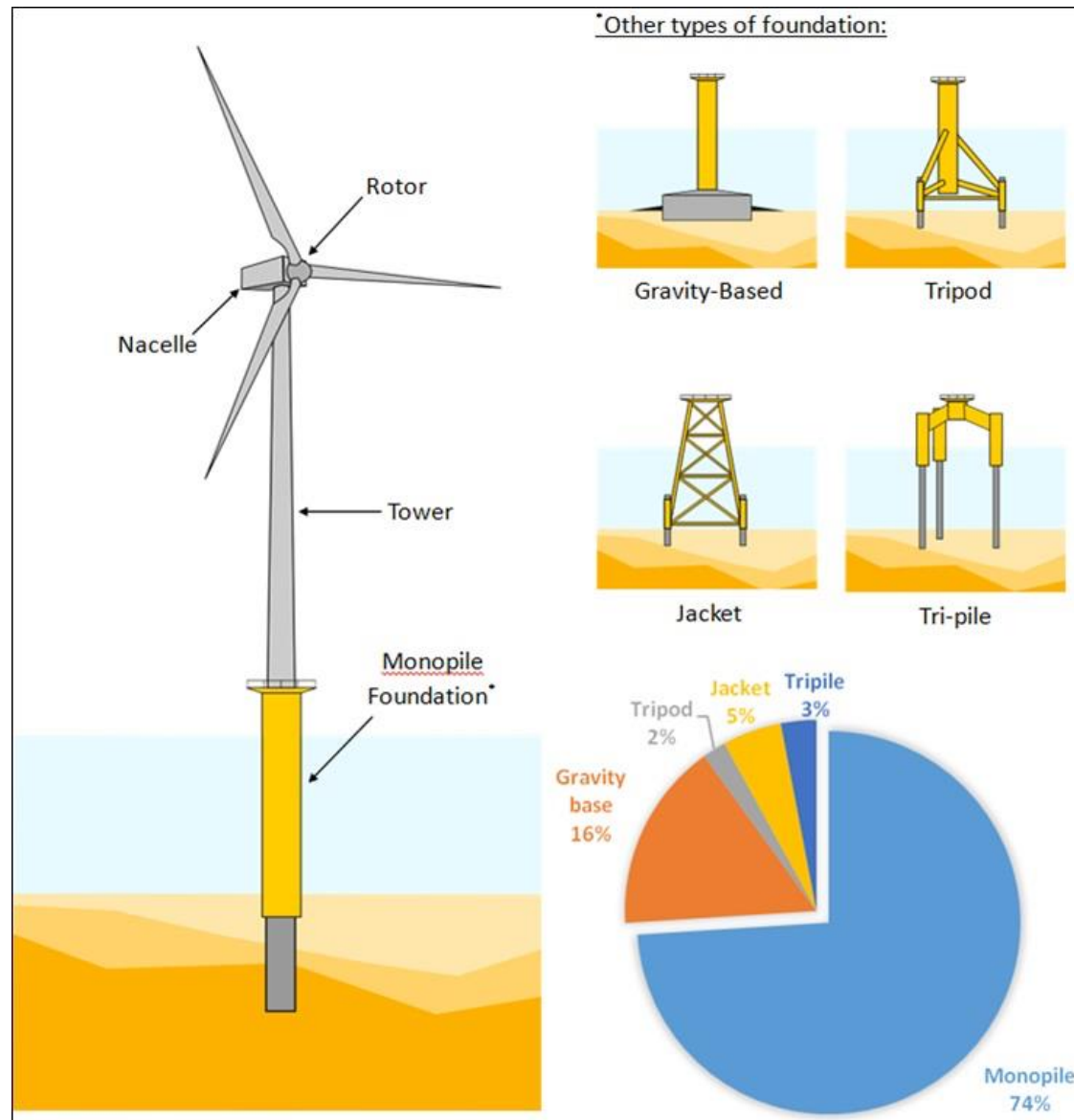


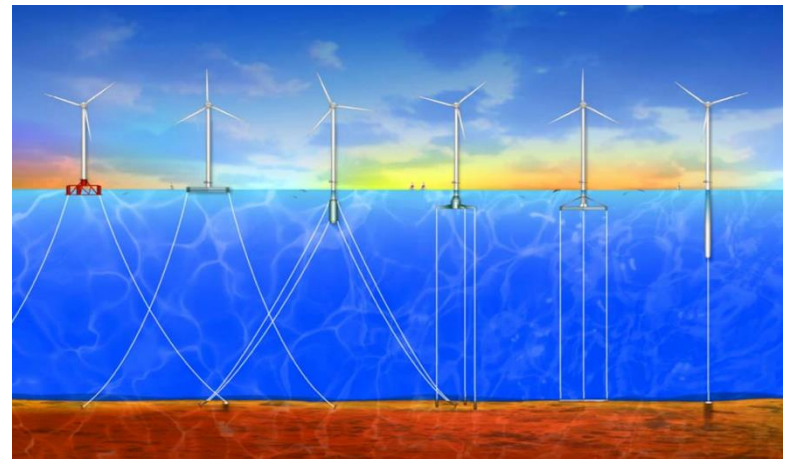
Figure source: VJ Tech

Foundation Tech (2/3)

- Transitional waves are waves traveling in water where depth is less than half the wavelength but greater than one-twentieth the wavelength ($1/20 L < D < 1/2 L$).
- Transitional waves are often wind-generated waves that have moved into shallower water.

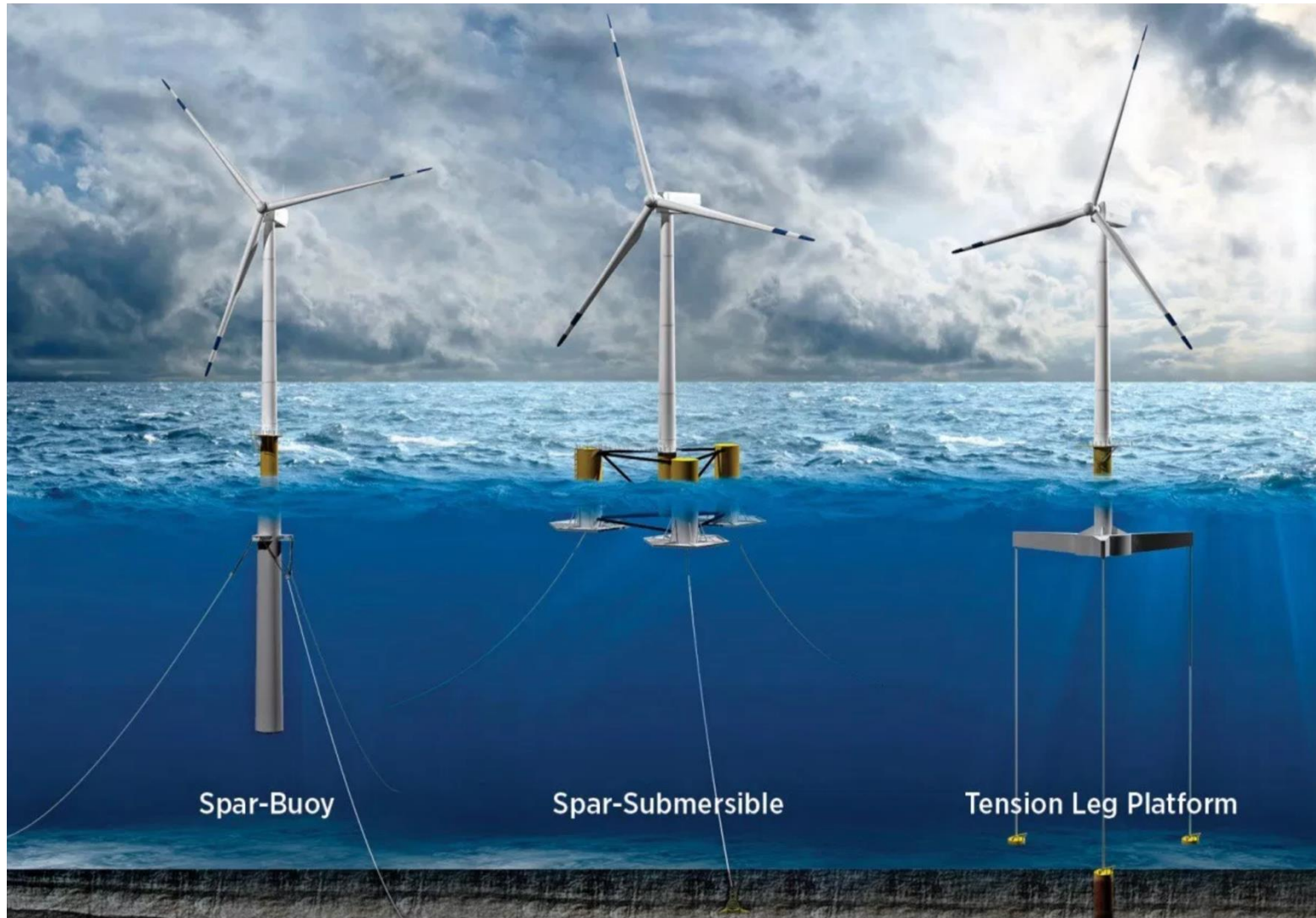


(a) Transitional depth foundations (30-60m)



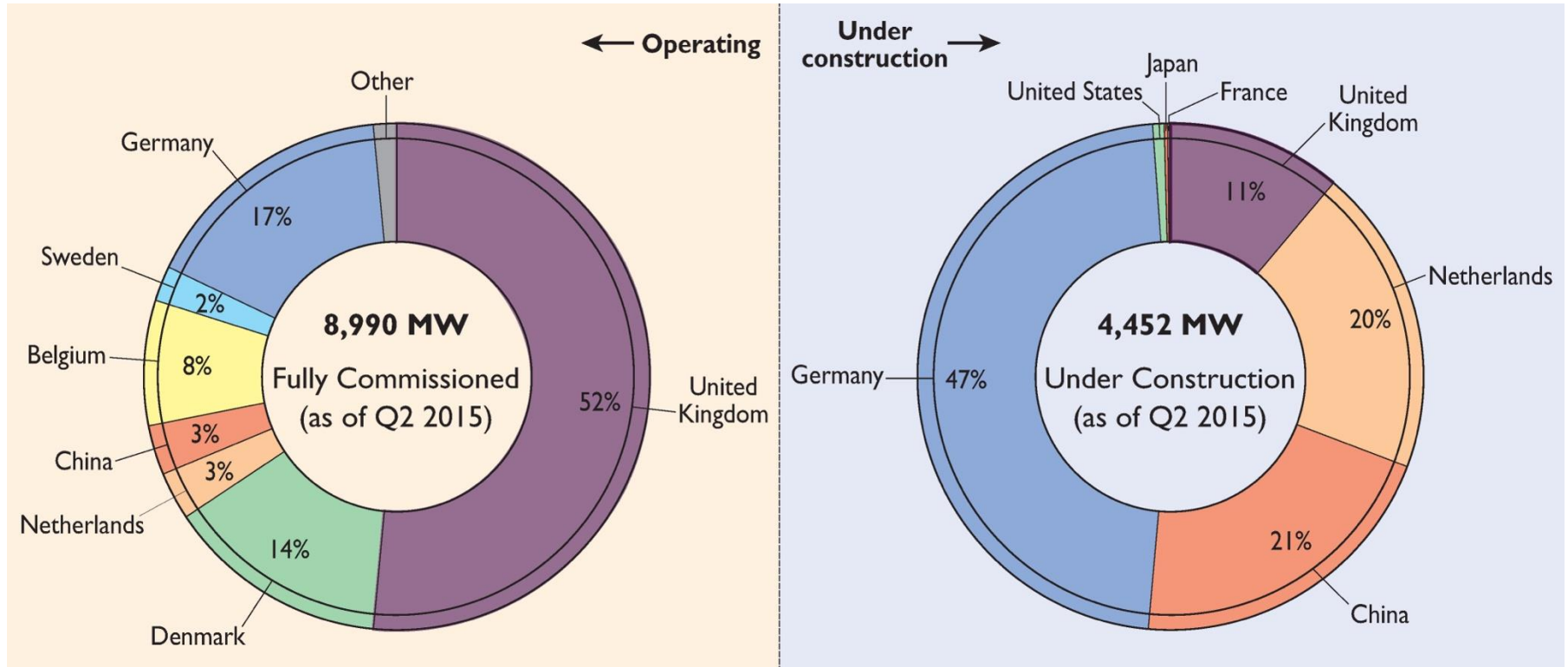
(b) Floating foundations (50-200m)

Foundation Tech (3/3)

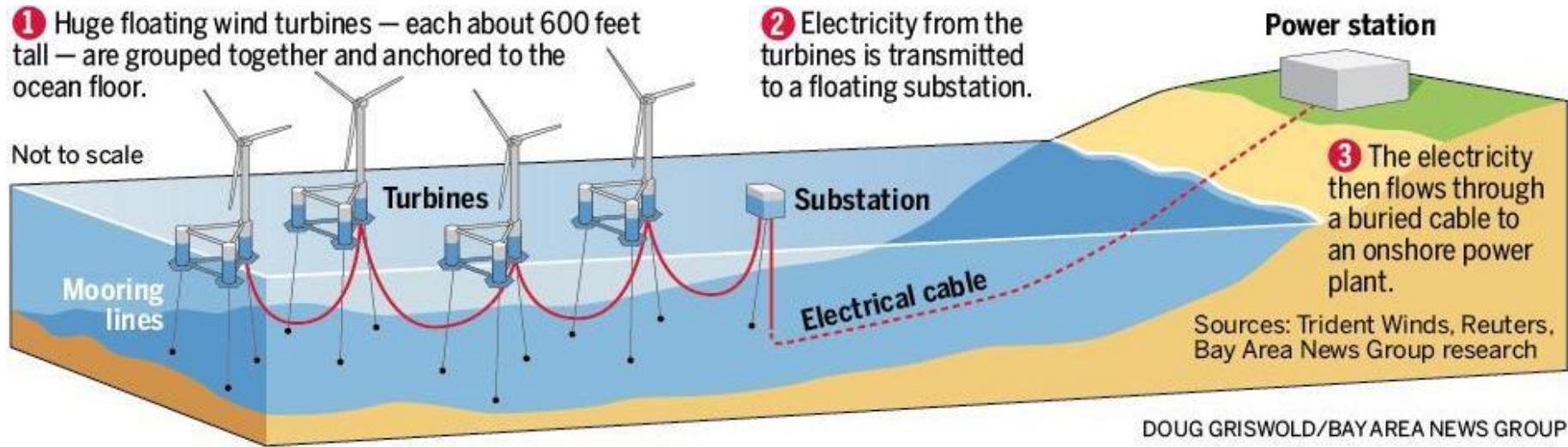


Floating Foundations, By Joshua Bauer, NREL

Global Offshore Wind Power Capacity



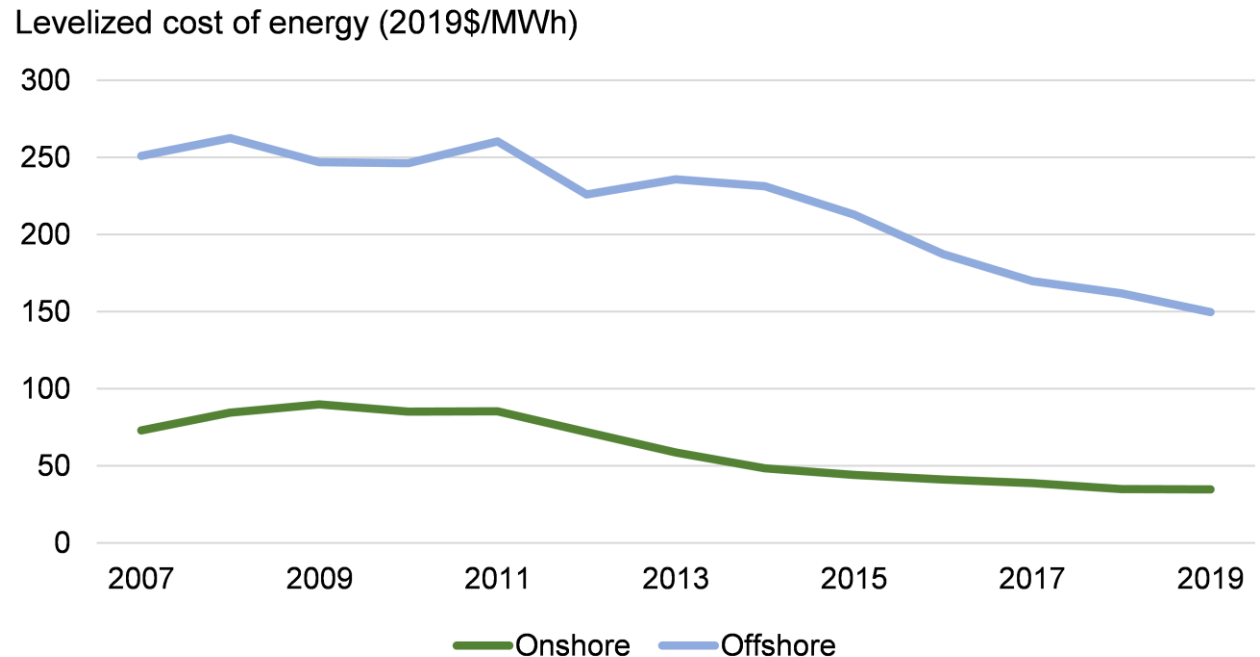
How Offshore Wind Farms Work



Electricity Cost from Wind

The costs of electricity from wind depend on:

- Annual energy production from the wind turbine installation
- Capital cost of the installation
- Discount rate being applied to the project
- Length of the contract with the purchaser
- Loan repaid
- O&M costs



Capital Cost Breakdown

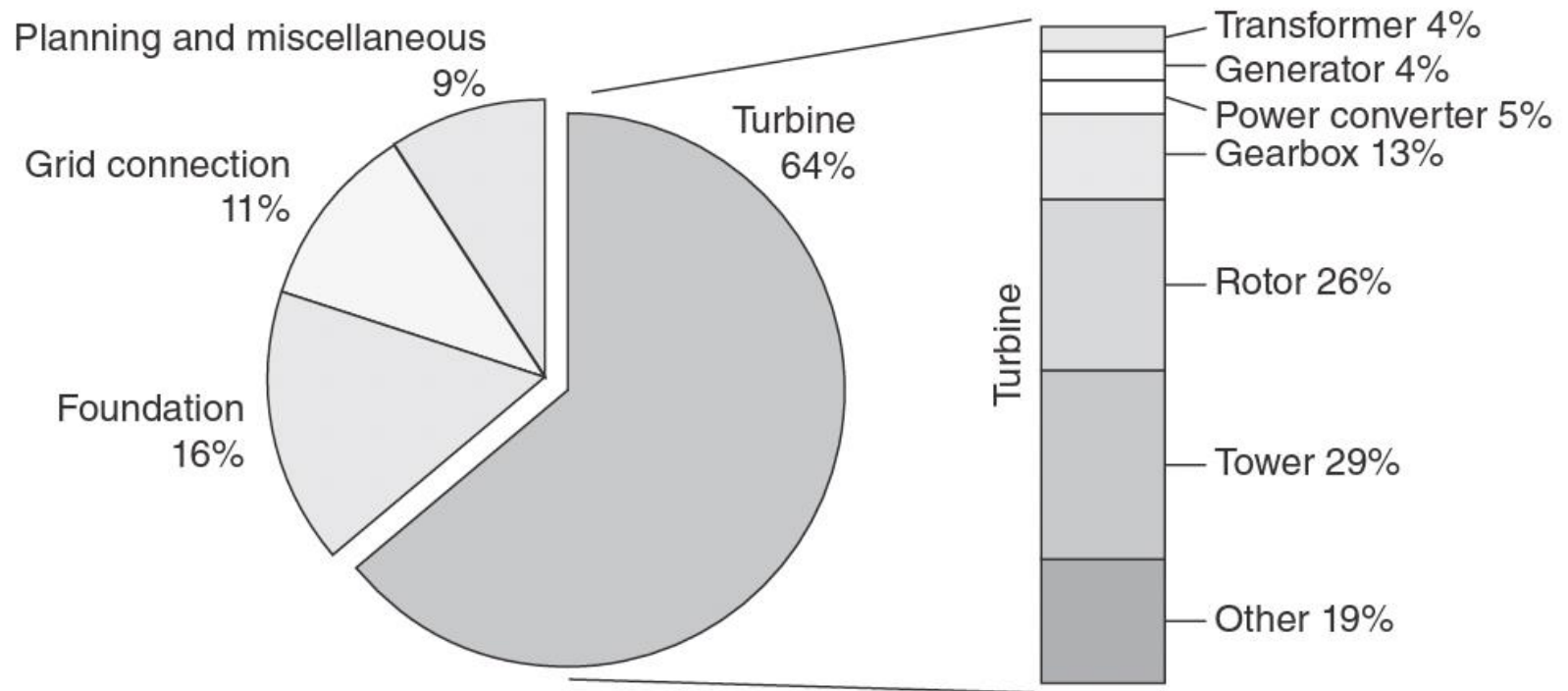


FIGURE 7.47 A breakdown of the capital costs of a wind system. Based on Blanco (2009) and EWEA (2007).

Capital Cost and Annual Cost

Table: An Example Cost Analysis for a 60-MW Wind Park

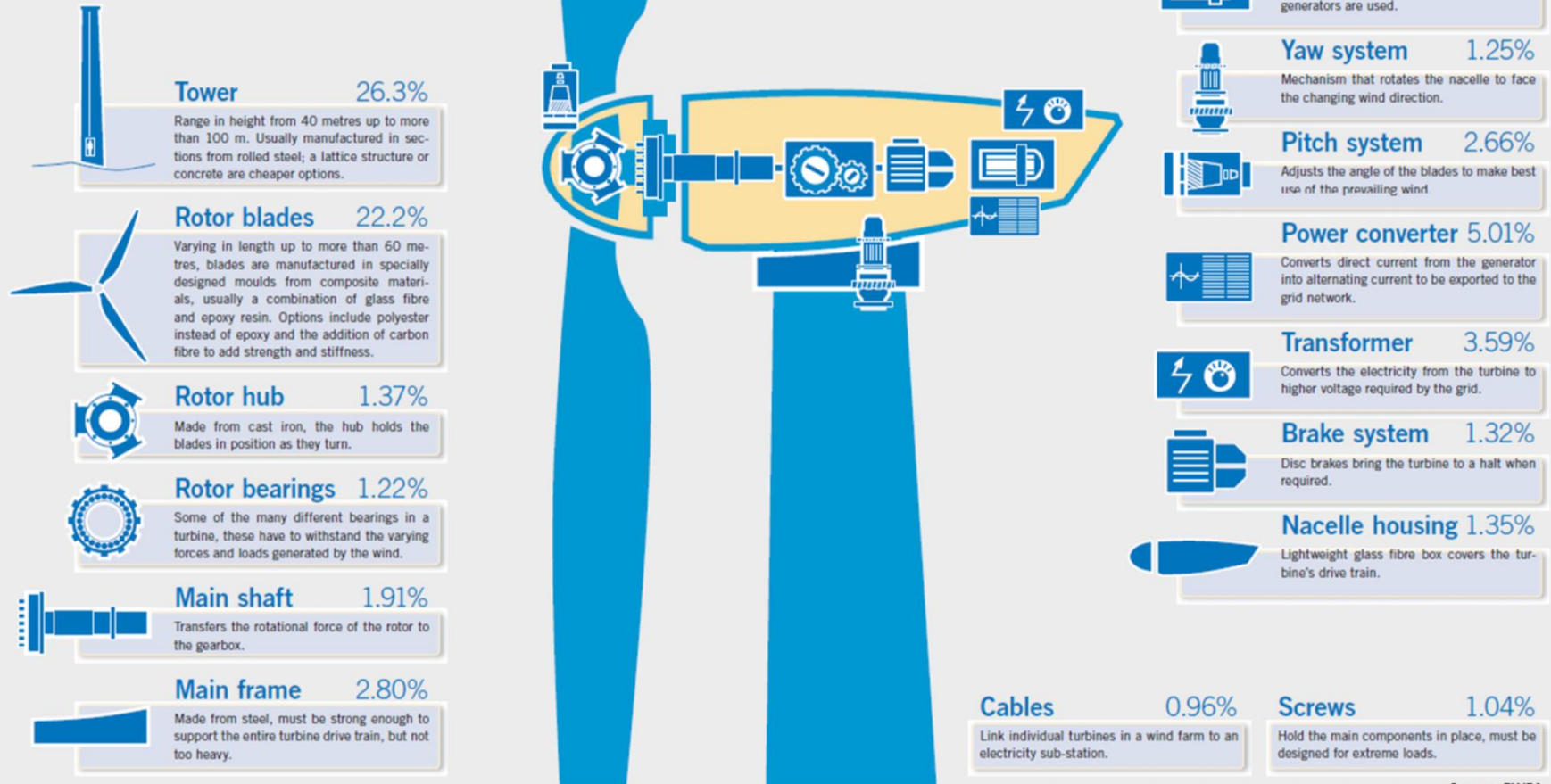
Capital Costs	Amount (\$)	Percentage
40 1.5-MW turbines @ \$1.1 M, spare parts	46,640,000	76.6
Site prep, grid connections	9,148,000	15.0
Interest during construction, contingencies	3,514,000	5.8
Project development, feasibility study	965,000	1.6
Engineering	611,000	1.0
Total Capital Cost	60,878,000	100.0
Annual Costs	Amount (\$/yr)	Percentage
Parts and labor	1,381,000	70.3
Insurance	135,000	6.9
Contingencies	100,000	5.1
Land lease	90,000	4.6
Property taxes	68,000	3.5
Transmission line maintenance	80,000	4.1
General and miscellaneous	111,000	5.6
Total Annual Costs	1,965,000	100.0

Source: Ministry of Natural Resources, Canada.

Cost Structure

How a wind turbine comes together

A typical wind turbine will contain up to 8,000 different components. This guide shows the main parts and their contribution in percentage terms to the overall cost. Figures are based on a REpower MM92 turbine with 45.3 metre length blades and a 100 metre tower.



Source: EWEA

Environmental Impact

Pros

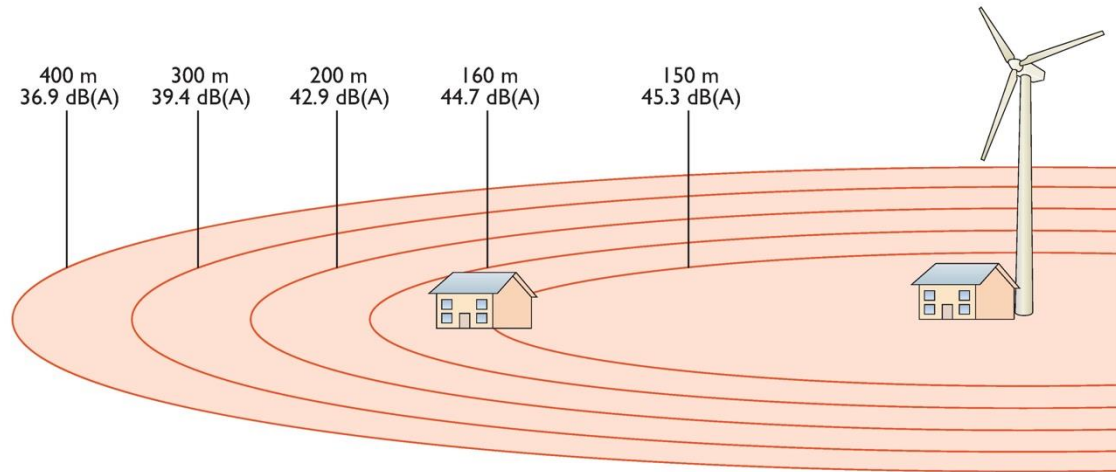
Generation of electricity by wind turbines does NOT involve:

- release of CO₂ or other greenhouse gases
- pollutants leading to acid rain or smog causing diseases
- radioactivity
- contamination of land, sea or water courses
- the consumption of water

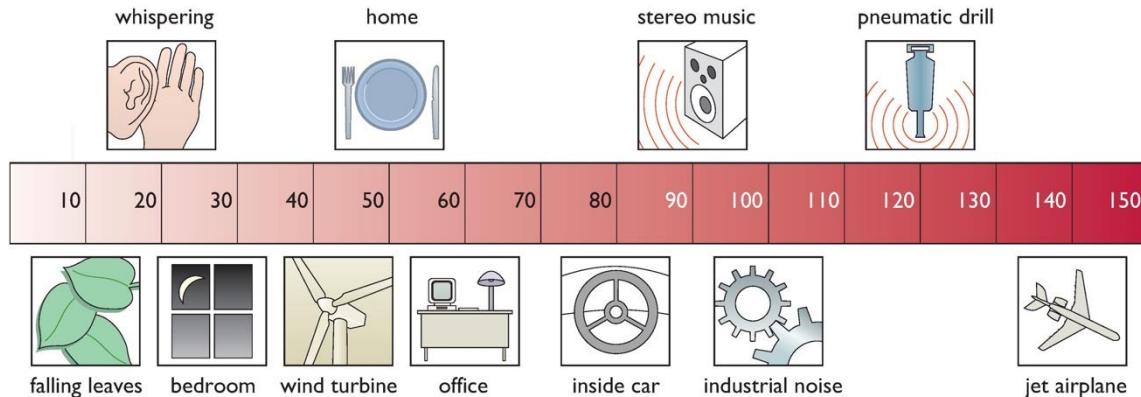
Cons

- noise
- electromagnetic interference
- aviation-related issues
- wildlife
- public attitudes and planning

Noise



Relative noise levels dB(A)

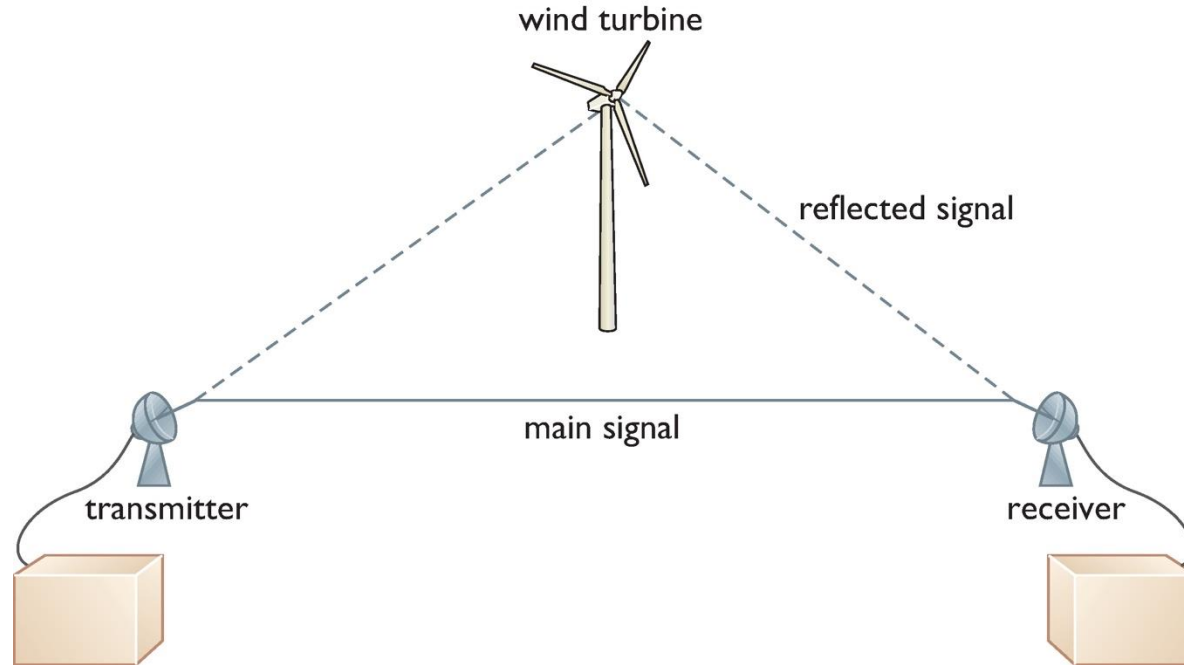


Mechanical noise

produced by equipment such as the gearbox

Aerodynamic noise due to the interaction of the airflow with the rotor

Electromagnetic Interference



Wind turbines can reflect some of the electromagnetic radiation: cause the received signal to be distorted significantly.

The extent of electromagnetic interference depends mainly on

- 1) the materials used to make the blades
- 2) the surface shape of the tower

Wind Turbines and Aviation



Solutions:

- 1) adapting the design of wind turbine blades to include radar absorbing materials.
- 2) development of systems that can filter out; e.g., BAE's advanced digital tracking system (Butler, 2007).

Impact on Wildlife

American Bird Conservancy, 2011

Obstruction	Collisions per year
wind turbines	100,000–440,000
towers	4–50 million
power lines	10–154 million
roads/vehicles	10.7–380 million
urban lights	31 million
glass on buildings	100 million–1 billion

Impact on Wildlife (Cont'd)



- Install radar systems that automatically detect approaching birds/bats.
- If there is a likelihood of collisions, bird deterrent devices can be activated or the turbines shut down until after the birds have passed.
- Help planners and wind farm operators take account of potential impacts on wildlife when developing and assessing wind turbines.



[Video: Meet the Dog Protecting Planes From Bird Strikes](#)

Public Attitudes and Planning Considerations

The visual perception of a wind farm is determined by

- turbine size/design/color
- number of blades/ turbines
- layout of the wind farm
- extent to which moving rotor blades attract attention.

